StudySapiens Complete Chapter Notes

Chapter 12 — Human Skeletal and Muscular Systems

Complete Chapter Notes · Textbook topics converted into board-style long questions, with important conceptual short questions inside every topic · PECTAA 2025–26, Punjab English Medium

This chapter follows the textbook topics 12.1 to 12.3. Each topic is converted into board-style long questions, and each topic ends with conceptual short questions. Direct definitions are omitted.

Board pattern (Punjab 2026, new syllabus): Chapter 12 carries 2 MCQs, 4 short questions in Q.3 (the highest short-question weight), and part (b) of a long question in Q.8. Shorts are 2 marks; longs are 8 marks.

TOPIC 12.1 — BONES AND CARTILAGE

Textbook topic 12.1 · printed pages 255–266

Q.1 Describe the structure of bone, bone marrow and bone cells. How does bone develop?

Board long question · 8 marks

Structure of bone

Bones are connective tissue reinforced with calcium and specialized bone cells. The surface is covered by **periosteum**. Beneath it is **compact bone**, whose basic units are **Haversian systems**. Each Haversian system consists of concentric **lamellae** of mineralized matrix containing collagen and calcium phosphate crystals. **Lacunae** contain mature bone cells called **osteocytes**, connected by **canaliculi**. The central **Haversian canal** contains blood vessels, nerves and lymphatics. Beneath compact bone is **spongy bone**, a latticework that is light and strong.

Bone marrow

Red bone marrow produces red blood cells, platelets and white blood cells. **Yellow bone marrow** fills the shafts of long bones, stores fat and can convert to red marrow during severe blood loss.

Bone cells

Cell	Function
Osteoblasts	Bone-forming cells; secrete unmineralized matrix and become osteocytes when surrounded by matrix
Osteocytes	Mature bone cells; maintain bone tissue and regulate calcium release
Osteoclasts	Bone-resorbing cells; break down bone and release calcium and phosphate into blood

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Bone development (osteogenesis)

Long bones develop by replacement of cartilage with bone (**ossification**). The centre of cartilage calcifies, chondrocytes die, blood vessels introduce osteoblasts and osteoclasts, and bone tissue replaces cartilage. Some bones of the skull develop directly from connective tissue without a cartilage stage. Even after formation, bone continues to remodel through resorption by osteoclasts and formation by osteoblasts.

Q.2 Describe cartilage and its types. Differentiate between bone and cartilage.

Board long question · 8 marks

Cartilage is a flexible connective tissue surrounded by **perichondrium**. Its matrix contains collagen, elastin and proteoglycans. It has **no blood vessels**, so chondrocytes are supplied by diffusion and cartilage heals slowly.

Type	Location	Feature
Hyaline cartilage	Nose, trachea, articulating surfaces of joints	Most common; smooth and flexible
Fibrocartilage	Intervertebral discs, pubic symphysis	Resists high stress and tension
Elastic cartilage	External ear, epiglottis	Highly flexible

Feature	Bone	Cartilage
External covering	Periosteum	Perichondrium
Cells	Osteoblasts, osteocytes, osteoclasts	Chondrocytes
Matrix	Calcium crystals and collagen	Collagen and other fibres
Blood supply	Present	Absent
Repair	Can grow and repair throughout life	Limited repair due to no direct blood supply

Q.3 Describe the arrangement of bones in the human skeleton.

Board long question · 8 marks

The human skeleton has **206 bones** and is divided into **axial** and **appendicular** skeletons.

Axial skeleton

- **Skull:** 22 bones — 8 cranial bones and 14 facial bones.
- **Middle ear ossicles:** 6 bones (malleus, incus, stapes in each ear).

- **Hyoid bone:** Single bone at the base of the skull; does not articulate with other bones.
- **Vertebral column:** 33 vertebrae — 7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral and 4 fused coccygeal.
- **Rib cage:** 12 pairs of ribs and sternum. The first 7 pairs are **true ribs**, pairs 8–10 are **false ribs**, and pairs 11–12 are **floating ribs**.

Appendicular skeleton

It includes the bones of the limbs and the girdles that attach them to the axial skeleton: the pectoral girdle (clavicle and scapula) and the pelvic girdle (hip bones).

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Q.4 Describe the types of joints in the human body. How does the skeleton support bipedal posture?

Board long question · 8 marks

Joint type	Movement	Examples
Fibrous	Little or no movement	Sutures of skull
Cartilaginous	Limited movement	Pubic symphysis, intervertebral discs, first rib–sternum joint
Synovial	Free movement	Shoulder, hip, elbow, knee

Synovial joints have articular cartilage, a fibrous capsule, synovial membrane and synovial fluid. They include ball-and-socket, hinge, pivot, ellipsoidal, saddle and gliding joints.

Bipedal posture

The S-shaped vertebral column distributes weight; the short, broad pelvis stabilizes the torso; the inward-angled femur keeps the centre of mass over the feet; large buttock and calf muscles propel the body; the foot arch absorbs shock; and short, non-prehensile toes act as levers during walking.

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Important Conceptual Short Questions — Topic 12.1

S1. How does the structure of compact bone provide both strength and flexibility?

Calcium phosphate crystals provide rigidity, while collagen fibres provide flexibility, making bone strong yet slightly flexible.

S2. Why do osteocytes need canaliculi?

Canaliculi connect osteocytes to each other and to the Haversian canal, allowing nutrient and waste exchange in the hard matrix.

S3. How do osteoblasts and osteoclasts work together in bone remodelling?

Osteoclasts break down old bone and release minerals, while osteoblasts form new bone, maintaining bone strength and calcium balance.

S4. Why does cartilage heal more slowly than bone?

Cartilage lacks blood vessels, so chondrocytes receive nutrients by diffusion, limiting repair capacity.

S5. Why are the first seven pairs of ribs called true ribs?

They attach directly to the sternum through their own costal cartilages.

S6. How does the vertebral column support bipedal posture?

Its S-shaped curve distributes body weight evenly and helps maintain balance while standing and walking.

S7. Why are synovial joints freely movable?

They have articular cartilage, synovial fluid and a joint cavity that reduce friction and allow a wide range of movement.

S8. How does the pelvis adapt humans for bipedal locomotion?

It is shorter and broader, stabilizing the torso and supporting body weight on two legs.

TOPIC 12.2 — DISORDERS OF SKELETAL SYSTEM

Textbook topic 12.2 · printed pages 266–270

Q.5 Describe the major disorders of the skeletal system and injuries to joints.

Board long question · 8 marks

- **Disc slip (herniated disc):** The outer layer of an intervertebral disc ruptures, allowing the inner gel-like substance to press on nerves or spinal cord, causing pain, numbness and weakness.
- **Spondylosis:** Degeneration of vertebrae, discs, ligaments or cartilage, causing bone outgrowths and pressure on nerves; common in lumbar and cervical regions.
- **Sciatica:** Compression or irritation of the sciatic nerve, often by a herniated disc, causing pain from lower back down the leg.
- **Arthritis:** Inflammation of joints. **Osteoarthritis** is degeneration of articular cartilage; **rheumatoid arthritis** is autoimmune inflammation of the synovial membrane; **gouty arthritis** is caused by uric acid crystal deposition.
- **Osteoporosis:** Decreased bone density making bones fragile, due to aging, menopause, calcium/vitamin D deficiency, lack of exercise, smoking or corticosteroids.

Injuries to joints

Dislocation is displacement of bones from their normal joint position. **Sprain** is stretching or tearing of ligaments. First aid includes rest, ice, compression and elevation, followed by medical assessment.

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Important Conceptual Short Questions — Topic 12.2

S1. How does a herniated disc cause pain and numbness?

The leaked inner disc material presses on nearby nerves or the spinal cord, disrupting normal nerve signals.

S2. Differentiate between osteoarthritis and rheumatoid arthritis.

Osteoarthritis is wear-and-tear degeneration of articular cartilage. Rheumatoid arthritis is an autoimmune inflammation of the synovial membrane.

S3. Why does gouty arthritis commonly affect the big toe?

Uric acid crystals tend to deposit in cooler peripheral joints, and the big toe joint is a common site of crystal-induced inflammation.

S4. How does osteoporosis increase fracture risk?

It reduces bone density and makes bones porous and fragile, so they break more easily under stress.

S5. Why is sciatica often linked to disc problems?

A herniated or bulging disc can compress the sciatic nerve, causing pain along its path from the lower back to the leg.

S6. Differentiate between dislocation and sprain.

Dislocation is displacement of bones from a joint, while sprain is stretching or tearing of ligaments supporting the joint.

TOPIC 12.3 — MUSCLES

Textbook topic 12.3 · printed pages 270–276

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Q.6 Compare skeletal, smooth and cardiac muscles.

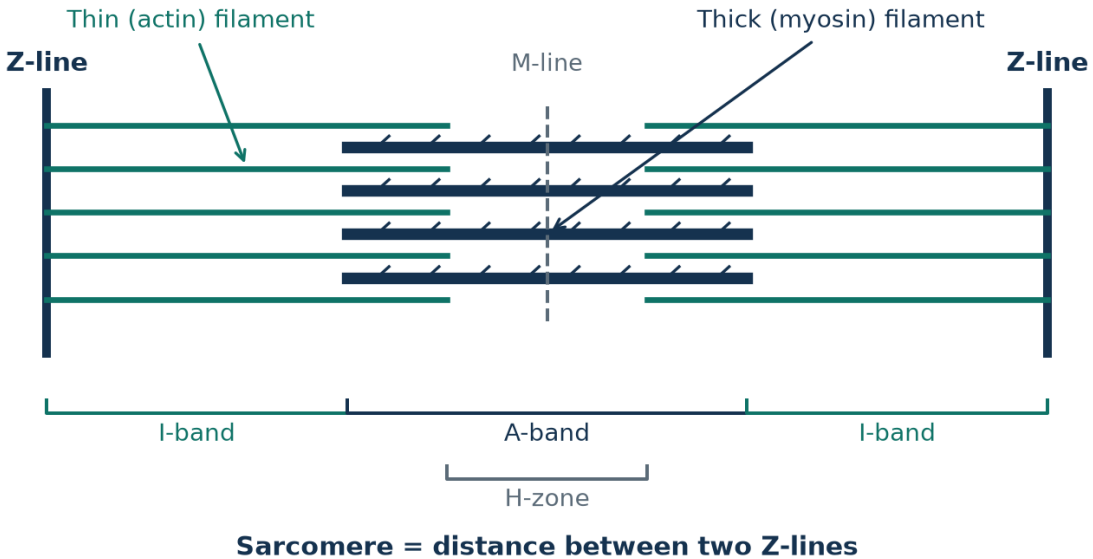
Board long question · 8 marks

Feature	Skeletal muscle	Smooth muscle	Cardiac muscle
Location	Attached to bones	Walls of internal organs and blood vessels	Heart wall only
Appearance	Striated	Non-striated	Striated and branched
Nuclei	Many per cell	One per cell	Usually one per cell
Control	Voluntary	Involuntary	Involuntary
Function	Moves skeleton	Moves substances through hollow organs	Pumps blood

Q.7 Describe the ultrastructure of skeletal muscle and explain the sliding filament model of muscle contraction.

Board long question · 8 marks

A skeletal muscle fibre is a cylindrical multinucleated cell enclosed by **sarcolemma**. Its cytoplasm, **sarcoplasm**, contains **sarcoplasmic reticulum** and **T-tubules**. Each fibre contains **myofibrils** made of thick **myosin** filaments and thin **actin** filaments. The region between two Z-lines is a **sarcomere**, the unit of contraction.



Structure of a sarcomere showing A-band, I-band, H-zone and Z-lines.

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Sliding filament model

1. A nerve impulse releases acetylcholine, triggering an impulse along the sarcolemma and T-tubules.
2. The sarcoplasmic reticulum releases Ca^{2+}, which binds to troponin and moves tropomyosin away from myosin-binding sites on actin.
3. Myosin heads bind actin, forming cross-bridges. Release of ADP and Pi causes the **power stroke**, pulling actin filaments toward the centre.
4. A new ATP molecule binds myosin, causing detachment. ATP hydrolysis re-cocks the head, and the cycle repeats.

During contraction, thin filaments slide deeper into the A-band, so the **H-zone and I-band narrow**, Z-lines come closer, and the sarcomere shortens. The A-band does not shorten.

Q.8 Explain antagonistic muscle action at the knee joint. Describe muscle fatigue, cramps and tetany.

Board long question · 8 marks

Skeletal muscles work in **antagonistic pairs**. At the knee, the **hamstrings** contract to flex (bend) the lower leg while the quadriceps relax. The **quadriceps** contract to extend (straighten) the leg while the hamstrings relax.

Muscle disorders

- **Muscle fatigue:** Decline in performance after prolonged/intense activity due to ATP depletion, accumulation of lactate/H^+/reactive oxygen, impaired calcium handling and micro-damage to fibres.
- **Muscle cramps:** Sudden, involuntary, painful contractions, often due to dehydration, salt imbalance, overuse or injury.
- **Tetany:** Involuntary spasms due to hyperexcitability of nerves, most commonly caused by **hypocalcaemia** (low blood calcium), vitamin D deficiency, renal failure or magnesium imbalance.

Tetany is different from tetanus. Tetany is a metabolic/electrolyte disorder, while tetanus is a bacterial infection causing sustained muscle spasm.

Important Conceptual Short Questions — Topic 12.3

S1. Why are skeletal muscles called voluntary while smooth and cardiac muscles are involuntary?

Skeletal muscle contractions are usually consciously controlled, whereas smooth and cardiac muscles contract without conscious control.

S2. How does the arrangement of actin and myosin produce striations in skeletal muscle?

Alternating dark A-bands (thick myosin filaments) and light I-bands (thin actin filaments) give skeletal muscle its striped appearance.

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S3. Why is calcium essential for muscle contraction?

It binds to troponin, moving tropomyosin away from myosin-binding sites on actin, allowing cross-bridge formation.

S4. How does ATP drive both contraction and relaxation of muscle?

ATP hydrolysis energizes the myosin head for the power stroke, and new ATP binding allows myosin to detach from actin for relaxation.

S5. Why do the H-zone and I-band narrow during contraction while the A-band remains constant?

Thin filaments slide deeper into the A-band, increasing overlap, but the length of thick filaments (A-band) does not change.

S6. How do antagonistic muscles produce movement at a joint?

One muscle contracts to move the bone in one direction while its opponent relaxes; reversal of contraction and relaxation moves the bone back.

S7. Why does ATP depletion cause muscle fatigue?

Without sufficient ATP, myosin heads cannot complete cross-bridge cycling and muscle contraction declines.

S8. How does hypocalcaemia cause tetany?

Low blood calcium increases nerve and muscle excitability, leading to involuntary spasms.

S9. Why does rigor mortis occur after death?

Cells can no longer produce ATP, so myosin heads cannot detach from actin, causing muscle stiffness.

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